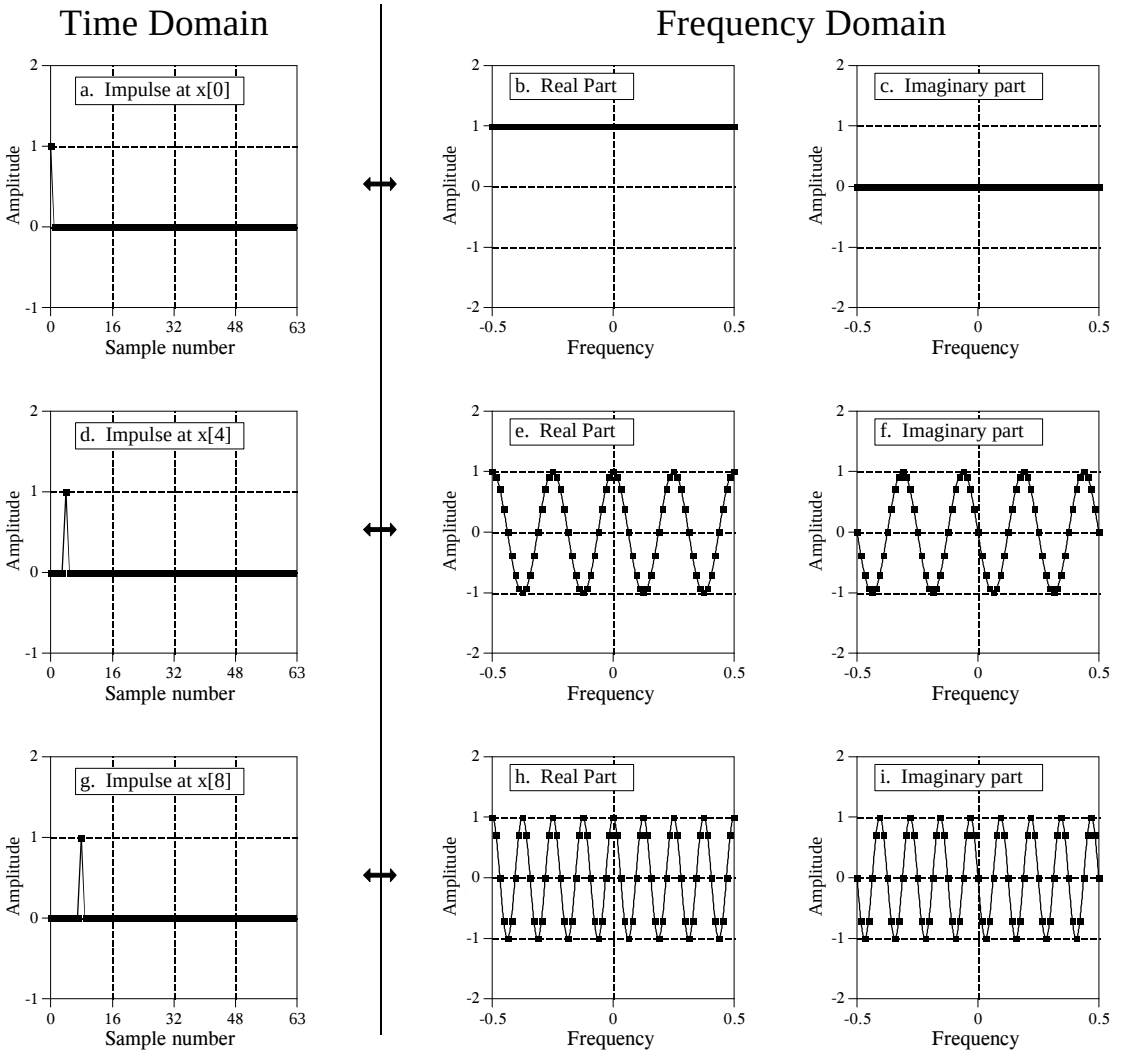

FIGURE 11-2
Delta function pairs in *rectangular form*. Each sample in the time domain results in a cosine wave in the real part, and a negative sine wave in the imaginary part of the frequency domain.

(f) show that an impulse at sample number four in the time domain results in four cycles of a cosine wave in the real part of the frequency spectrum, and four cycles of a negative sine wave in the imaginary part. As you recall, an impulse at sample number four in the real part of the frequency spectrum results in four cycles of a cosine wave in the time domain. Likewise, an impulse at sample number four in the imaginary part of the frequency spectrum results in four cycles of a negative sine wave being added to the time domain wave.

As mentioned in Chapter 8, this can be used as another way to calculate the DFT (besides correlating the time domain with sinusoids). Each sample in the time domain results in a cosine wave being added to the real part of the